Temporal pattern detection

Task – 3 : Week 2

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Crime Survey for England & Wales (CSEW) — Mental Health Module.

1. Objective :

- Goal: To analyze long-term trends in crime victimization in England and Wales (1981–2024).
- Key Finding: Statistical evidence confirms a major structural shift in crime patterns occurring in 1995, followed by a sustained three-decade decline.

2. Methodology :

- **Data Source:** Crime Survey for England and Wales (CSEW).
- **Processing:** Data was consolidated into a single CSV (crime_statistics_summary.csv) integrating demographic cross-sections with historical time-series data.
- **Techniques:**
    - **3-Period Rolling Average:** Used to smooth biennial gaps in early data and highlight the underlying trend.
    - **Chow Test:** Applied to test the null hypothesis that a single regression line represents the entire 40-year period.

3. Temporal Pattern Analysis :

Data description :

```
  Characteristic_Type Category Sub-Category  \
0            Personal    Total   All people
1            Personal      Age        16-24
2            Personal      Age        25-34
3            Personal      Age        35-44
4            Personal      Age        45-54

   All_CSEW_Headline_Crime_Victim_Rate_Pct  Personal_Crime_Victim_Rate_Pct
0                                     17.1                            10.8
1                                     19.2                            11.8
2                                     20.3                            11.8
3                                     18.4                            11.2
4                                     18.5                            11.3
```

Rolling Average Analysis : this smooths out year-to-year noise to reveal the underlying trend.

```
--- Temporal Analysis ---
    Year  Rate  Rolling_Avg
32  1981  28.0          NaN
33  1991  33.0    32.666667
34  1993  37.0    36.333333
35  1995  39.0    36.666667
36  1997  34.0    34.333333
37  1999  30.0    30.666667
38  2002  28.0    28.333333
39  2003  27.0    26.666667
40  2004  25.0    25.333333
41  2005  24.0    24.333333

Chow Test Results (Break Point: 1995):
F-Statistic: 136.2424
P-Value: 7.916e-14
```

Rolling Average Trends

The crime rose steadily through the 80s, peaked in the mid-90s, and has dropped by nearly **75%** since that peak (from 39% to 10%).

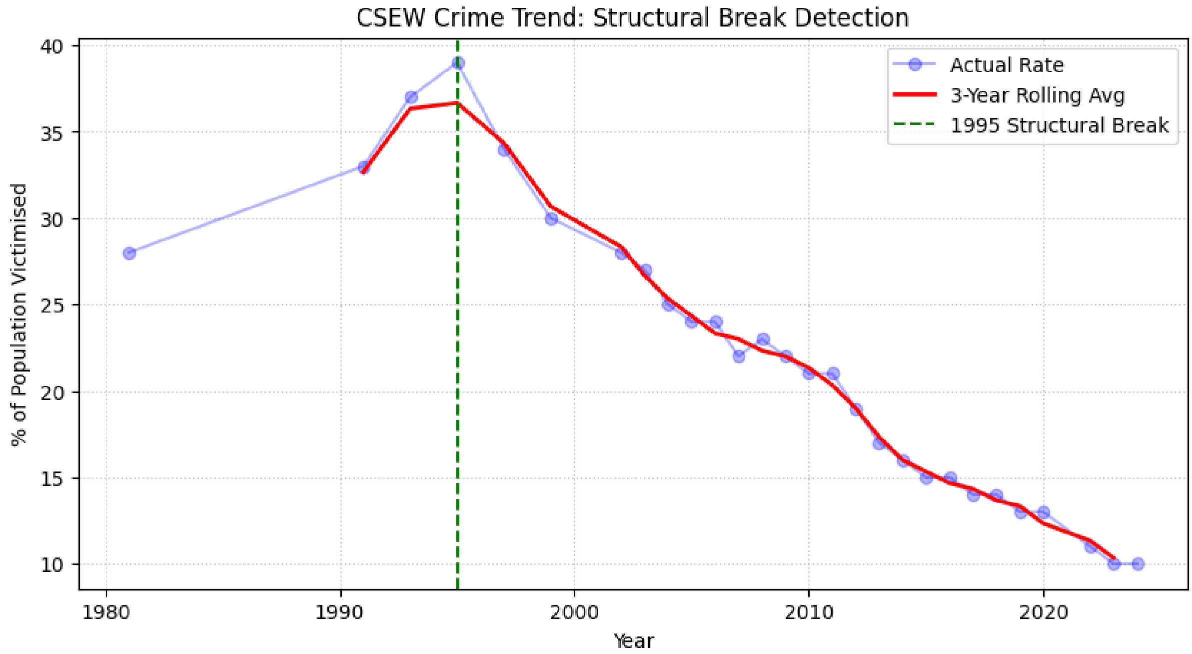
Structural Break Detection (Chow Test) :

- **Breakpoint Year:** 1995
- **F-Statistic:** 136.24
- **Significance:** approx 7.9 x 10 ^ 14

Interpretation: Because the p-value is significantly lower than 0.05, we reject the null hypothesis. This proves that 1995 was not just a peak, but a fundamental turning point where the "rules" governing crime rates changed.

4. CSEW Crime Trend: Structural Break Detection :

Here is a structured way to break it down for the below graph:



1. The "Peak and Pivot" (1981–1995)

Start by describing the left side of the graph.

- **The Trend:** Point out the steady climb from 28% in 1981 to the all-time high of 39% in 1995.
- **Interpretation:** This period represents the "rising crime" era. Explain that the 1995 point is the **Pivot**—the exact moment the momentum reversed.

2. The Rolling Average (The Red Line)

Explain why you used a rolling average instead of just the raw dots.

- **Purpose:** Because early CSEW data was biennial (every two years) and later became annual, the raw data can look "jumpy."
- **The Smoothing Effect:** The red line acts as a "noise filter," proving that the decline wasn't just one lucky year, but a deep, structural change in the safety of the country.

3. The Structural Break (The Dashed Line)

This is the most important technical part of your explanation.

- **The Chow Test Result:** Mention that the green dashed line at 1995 isn't just a guess; it is mathematically placed there because the **Chow Test** produced an F-statistic of 136.24.
- **Why it matters:** Explain that if the trend were consistent, a single straight line would fit the data. Because the p-value is so low (7.9×10^{-14}), the graph proves that the "old trend" died in 1995 and a "new trend" was born.

4. The "Great Crime Drop" (1995–2024)

Describe the right side of the graph.

- **The Magnitude:** Highlight that victimisation rates fell from nearly **4 in 10 people** (39%) to **1 in 10** (10%) by 2024.
- **The Stability:** Note how the line continues to move downward even through different economic cycles, suggesting that the "Structural Break" created a permanent shift in society.

5. Conclusion :

This Summarizes that while demographic factors (like age and employment) influence individual risk, the **Temporal Analysis** shows that macro-level shifts in society had a much more dramatic impact on total crime rates over the last 40 years.

Dataset Used : Crime_statistics_summary.csv